Constant Maturity Asset Swap Convexity Correction

Working notes on Pucci (2012a)

Reference paper: M. Pucci, Constant Maturity Asset Swap Convexity Correction*,*
Banca IMI, Milan, February 27, 2012. SSRN working paper no. 1961545.
Available at https://ssrn.com/abstract=1961545.

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1 Trade and Motivation

An *asset swap* (ASW) is a package consisting of a discount bond purchased at par plus a payer swap exchanging the bond's (non-credit) coupons for Libor plus a spread R^{asw}. The spread is set to make the package fair: it compensates the investor for buying-at-par a bond trading below par because of credit risk or coupon mismatch.

A *Constant Maturity Asset Swap* (CMASW) is a swap leg indexed to the prevailing ASW spread of a specific reference bond – the natural analogue of a CMS swap, with the rolling rate replaced by the spread. The elemental building block is the *ASW-let*: a single coupon paying, at T_p,

$$\text{coverage} \cdot R^{\text{asw}}_{T_0}$$

i.e. the asset swap spread observed at T_0 on the reference bond.

Use case. An investor with a 3Y rolling 5Y duration position in Italian government bonds enters a 3Y swap paying fixed and receiving a floating CMASW. A deterioration of Italian credit (which marks the bond portfolio down) is then offset by the rising floating leg.

Modelling goal. Derive the convexity correction for the ASW-let in closed form, capturing the (cross-)gamma cost incurred when hedging via dynamic trading of the forward asset-swap package.

2 Definitions and Setup

2.1 Schedule and bond curves

Let $T_0 < T_1 < \cdots < T_n$ be a payment schedule (the bond's coupon schedule, also used for the asset swap leg up to immaterial mismatches), $y_i = \text{yf}(T_{i-1}, T_i)$ the year fractions, D_{tT} the risk-free zero-coupon bond, and A_t the risk-free annuity:

$$A_t \;\equiv\; A_t(T_1, \ldots, T_n) \;=\; \sum_{i=1}^{n} y_i \, D_{tT_i}. \tag{1}$$

Denote B_t the price process of the reference bond and B_t^{rf} its risk-free counterpart (same coupon structure, discounted on the benchmark curve).

2.2 Forward asset swap spread

Definition 1 (Forward ASW spread)**.** *For all $t \le T_0$,*

$$R^{\text{asw}}_{tT_0} \;\equiv\; \frac{B^{\text{rf}}_t - B_t}{A_t}. \tag{2}$$

This generalises the textbook ASW spread in three useful directions:

1. the package may be *forward starting* ($T_0 > 0$);
2. B^{rf}_t can be replaced by any benchmark bond (allowing flow mismatches);
3. the definition survives a credit event on the reference bond – B_t may default to a recovery process.

Lemma 1 (ASW spread is a martingale under the swaption measure)**.** *The forward ASW spread $R^{\text{asw}}_{tT_0}$ is a martingale under the A_t-numeraire (swaption) measure $\mathbb{Q}^A$.*

Proof. Both B^{rf}_t and B_t are tradable assets, hence B^{rf}_t/A_t and B_t/A_t are $\mathbb{Q}^A$-martingales. The difference is therefore a martingale, and $R^{\text{asw}}_{tT_0} = (B^{\text{rf}}_t/A_t) - (B_t/A_t)$. □

Remark 1 (Sign of the ASW spread)**.** *While R^{swp} is naturally non-negative, R^{asw} may be* negative *("premium bonds" – historically the case for treasuries when interbank deposits were perceived less risk-free than government). This motivates the displaced-diffusion specification adopted below.*

For the rest of the note, T_0 is fixed and we drop the postfix: $R^{\text{asw}}_t \equiv R^{\text{asw}}_{tT_0}$, $R^{\text{swp}}_t \equiv R^{\text{swp}}_{tT_0}$.

3 Pricing Framework

3.1 Generic pricing formula

For payment date $T_p \ge T_0$ (most commonly $T_p = T_1$), the ASW-let's arbitrage-free price is, under any equivalent martingale measure $\mathbb{Q}^N$ with numeraire N_t,

$$v_0 \;=\; N_0\, \mathbb{E}^N_0\left[\frac{R^{\text{asw}}_{T_0}\, D_{T_0T_p}}{N_{T_0}}\right]. \tag{3}$$

3.2 Convexity correction

In line with the CMS convention, we focus on the convexity correction rather than the bare price:

Definition 2 (Convexity correction)**.**

$$\text{CC} \;\equiv\; \frac{N_0}{D_{0T_p}}\, \mathbb{E}^N_0\left[\frac{R^{\text{asw}}_{T_0}\, D_{T_0T_p}}{N_{T_0}}\right] - R^{\text{asw}}_0, \tag{4}$$

so that the ASW-let's normalised value reads $D_{0T_p}\big(R^{\text{asw}}_0 + \text{CC}\big)$.

By Lemma 1 we pick $N_t = A_t$ (swaption measure), making R^{asw}_t a martingale and giving the working form

$$\text{CC} \;=\; \frac{A_0}{D_{0T_p}}\, \mathbb{E}^A_0\left[R^{\text{asw}}_{T_0} \cdot \frac{D_{T_0T_p}}{A_{T_0}}\right] - R^{\text{asw}}_0. \tag{5}$$

The numerator inside the expectation is a function of the T_0-information only – the modelling task is to specify the joint distribution of $R^{\text{asw}}_{T_0}$ with the (unknown) ratio $D_{T_0T_p}/A_{T_0}$.

4 One-Factor Linear Swap-Rate Model

4.1 Model assumption

Assumption 1 (Linear swap-rate model)**.** *There exist functions G_U such that, for every $U \in \{T_p, T_1, \ldots, T_n\}$,*

$$\frac{D_{T_0 U}}{A_{T_0}} \;=\; G_U\big(R^{\mathrm{swp}}_{T_0}\big), \qquad G_U(x) \;=\; \alpha + \beta_U\, x. \tag{6}$$

This is the same structural assumption Hagan (2003) uses for CMS pricing and Pelsser (2001), Boenkost–Schmidt (2003) and Zheng–Kwok (2010) justify under various practical setups.

4.2 Calibration

Proposition 1 (Calibration of α, β_U)**.** *Under Assumption 1,*

$$\alpha \;=\; \frac{1}{\sum_{i=1}^{n} y_i}, \qquad \beta_U \;=\; \frac{D_{0U}/A_0 - \alpha}{R^{\mathrm{swp}}_0}. \tag{7}$$

Proof. For *any* R^{swp}_0, the algebraic identity $\sum_{i=1}^{n} y_i D_{0T_i} = A_0$ rearranges to

$$1 \;=\; \sum_{i=1}^{n} y_i\, \frac{D_{0T_i}}{A_0} \;=\; \alpha\left(\sum_{i=1}^{n} y_i\right) + R^{\mathrm{swp}}_0 \sum_{i=1}^{n} y_i\, \beta_{T_i}, \tag{8}$$

where the second equality uses $\mathbb{E}^A[D_{T_0 T_i}/A_{T_0} \mid \mathcal{F}_0] = D_{0T_i}/A_0$ together with (6). Sending $R^{\mathrm{swp}}_0 \to 0$ in (8) yields $\alpha = 1/\sum y_i$. The expression for β_U then follows from the martingale property

$$\alpha + \beta_U\, R^{\mathrm{swp}}_0 \;=\; \mathbb{E}^A_0\left[\frac{D_{T_0 U}}{A_{T_0}}\right] \;=\; \frac{D_{0U}}{A_0},$$

solved for β_U. □

Remark 2 (Multi-curve compatible)**.** *Equation (6) only assumes that the discount-to-annuity ratio depends on a single state variable – the framework extends to multi-curve setups where the forward swap rate is specified independently of the discount/funding factors used to compute A_t.*

4.3 Main result

Proposition 2 (Linear-Swap ASW Convexity Correction)**.** *Under Assumption 1 and Proposition 1,*

$$\boxed{\mathrm{CC} \;=\; \left(1 - \frac{A_0\,\alpha}{D_{0T_p}}\right)\left(\frac{\mathbb{E}^A_0\big[R^{\mathrm{asw}}_{T_0}\, R^{\mathrm{swp}}_{T_0}\big]}{R^{\mathrm{swp}}_0} - R^{\mathrm{asw}}_0\right).} \tag{9}$$

Proof. Substitute the linear form (6) for $U = T_p$ into the working CC expression (5):

$$\begin{aligned}
\mathrm{CC} &= \frac{A_0}{D_{0T_p}}\, \mathbb{E}^A_0\Big[R^{\mathrm{asw}}_{T_0}\big(\alpha + \beta_{T_p} R^{\mathrm{swp}}_{T_0}\big)\Big] - R^{\mathrm{asw}}_0 \\
&= \frac{A_0\,\alpha}{D_{0T_p}}\, \mathbb{E}^A_0\big[R^{\mathrm{asw}}_{T_0}\big] + \frac{A_0\,\beta_{T_p}}{D_{0T_p}}\, \mathbb{E}^A_0\Big[R^{\mathrm{asw}}_{T_0} R^{\mathrm{swp}}_{T_0}\Big] - R^{\mathrm{asw}}_0.
\end{aligned}$$

Use $\mathbb{E}_0^A[R_{T_0}^{\text{asw}}] = R_0^{\text{asw}}$ (martingale) and substitute $\beta_{T_p} = (D_{0T_p}/A_0 - \alpha)/R_0^{\text{swp}}$ from (7):

$$\begin{aligned}
\text{CC} &= \frac{A_0\alpha}{D_{0T_p}} R_0^{\text{asw}} + \frac{D_{0T_p}/A_0 - \alpha}{R_0^{\text{swp}}} \cdot \frac{A_0}{D_{0T_p}} \mathbb{E}_0^A\left[R_{T_0}^{\text{asw}} R_{T_0}^{\text{swp}}\right] - R_0^{\text{asw}} \\
&= \left(\frac{A_0\alpha}{D_{0T_p}} - 1\right) R_0^{\text{asw}} + \frac{D_{0T_p} - A_0\alpha}{R_0^{\text{swp}}\, D_{0T_p}} \mathbb{E}_0^A\left[R_{T_0}^{\text{asw}} R_{T_0}^{\text{swp}}\right] \\
&= \left(1 - \frac{A_0\alpha}{D_{0T_p}}\right)\left(\frac{\mathbb{E}_0^A\left[R_{T_0}^{\text{asw}} R_{T_0}^{\text{swp}}\right]}{R_0^{\text{swp}}} - R_0^{\text{asw}}\right).
\end{aligned}$$

□

The structure is transparent: a deterministic prefactor controlled by $\alpha A_0/D_{0T_p}$ multiplies a *cross-moment* term – this is what carries the (cross-)gamma cost of dynamically replicating the forward asset-swap package.

Sign rule. For payment date T_p close to T_1 (the most common choice), $D_{0T_p} > A_0\alpha$, so the prefactor $1 - A_0\alpha/D_{0T_p}$ is positive. The CC then has the sign of $\mathbb{E}[R_{T_0}^{\text{asw}} R_{T_0}^{\text{swp}}] - R_0^{\text{asw}} R_0^{\text{swp}}$, i.e. the sign of the covariance.

5 Closed-Form Distributions

5.1 Displaced-lognormal dynamics

Specify both rates as displaced lognormal under $\mathbb{Q}^A$, with displacements $a_{\text{swp}}, a_{\text{asw}}$ and instantaneous correlation ρ:

$$\frac{dR_t^{\text{swp}}}{R_t^{\text{swp}} + a_{\text{swp}}} = \sigma_{\text{swp}}\, dW_t^{\text{swp}}, \tag{10}$$

$$\frac{dR_t^{\text{asw}}}{R_t^{\text{asw}} + a_{\text{asw}}} = \sigma_{\text{asw}}\, dW_t^{\text{asw}}, \tag{11}$$

$$d\langle W^{\text{swp}}, W^{\text{asw}}\rangle_t = \rho\, dt. \tag{12}$$

Remark 3 (Attainability)**.** *For each underlying we need either $R_0 > -a$ (*lognormal *regime, support $(-a, \infty)$) or $0 < R_0 < -a$ (*anti-lognormal*, support $(-\infty, -a)$). Asset swap spreads can be negative ("premium bonds"), motivating the displaced choice with $a < 0$ when appropriate.*

5.2 Cross-moment closed form

Lemma 2 (Cross-moment)**.** *Under* (10)–(12)*,*

$$\begin{aligned}
\mathbb{E}_0^A\big[R_{T_0}^{\text{swp}}\, R_{T_0}^{\text{asw}}\big] = (R_0^{\text{swp}} + a_{\text{swp}})(R_0^{\text{asw}} + a_{\text{asw}}) \exp\big(\sigma_{\text{swp}}\sigma_{\text{asw}}\rho\, T_0\big) \\
-\ a_{\text{swp}}(R_0^{\text{asw}} + a_{\text{asw}})\ -\ a_{\text{asw}}(R_0^{\text{swp}} + a_{\text{swp}})\ +\ a_{\text{swp}}a_{\text{asw}}.
\end{aligned} \tag{13}$$

Proof. Define the shifted variables $X_t = R_t^{\text{swp}} + a_{\text{swp}}$, $Y_t = R_t^{\text{asw}} + a_{\text{asw}}$. Then X_t, Y_t are positive (under the lognormal regime) $\mathbb{Q}^A$-martingales solving

$$\frac{dX_t}{X_t} = \sigma_{\text{swp}}\, dW_t^{\text{swp}}, \qquad \frac{dY_t}{Y_t} = \sigma_{\text{asw}}\, dW_t^{\text{asw}},$$

so $\log X_T$ and $\log Y_T$ are jointly Gaussian with covariance $\sigma_{\text{swp}}\sigma_{\text{asw}}\rho\, T_0$. By the standard moment generating function of a bivariate lognormal:

$$\mathbb{E}_0^A[X_{T_0}\, Y_{T_0}] = X_0\, Y_0 \exp(\sigma_{\text{swp}}\sigma_{\text{asw}}\rho\, T_0).$$

Expanding $R_{T_0}^{\text{swp}} R_{T_0}^{\text{asw}} = (X_{T_0} - a_{\text{swp}})(Y_{T_0} - a_{\text{asw}})$ and using the martingale property $\mathbb{E}[X_{T_0}] = X_0$, $\mathbb{E}[Y_{T_0}] = Y_0$ gives the result. □

Substituting Lemma 2 into Proposition 2 yields a fully closed-form CC.

5.3 Lognormal limit

Corollary 1 (Lognormal CC)**.** *With* $a_{\text{swp}} = a_{\text{asw}} = 0$,

$$\boxed{\text{CC} \;=\; R_0^{\text{asw}}\left(1 - \frac{A_0\,\alpha}{D_{0T_p}}\right)\left(e^{\sigma_{\text{swp}}\sigma_{\text{asw}}\rho\,T_0} - 1\right).} \tag{14}$$

Proof. With $a_{\text{swp}} = a_{\text{asw}} = 0$, (13) reduces to $\mathbb{E}[R_{T_0}^{\text{swp}} R_{T_0}^{\text{asw}}] = R_0^{\text{swp}} R_0^{\text{asw}} \exp(\sigma_{\text{swp}}\sigma_{\text{asw}}\rho T_0)$, so (9) becomes $(1 - A_0\alpha/D_{0T_p})\big(R_0^{\text{asw}} e^{\sigma_{\text{swp}}\sigma_{\text{asw}}\rho T_0} - R_0^{\text{asw}}\big)$. □

Limits.

1. $\rho = 0$ **(lognormal).** CC $= 0$. Independence kills the cross-gamma in the lognormal case, but *not* in the displaced case because of the additive constants in (13).
2. $\sigma_{\text{swp}} = 0$ **or** $\sigma_{\text{asw}} = 0$**.** CC $= 0$ in the lognormal case – one of the two rates is frozen at its initial value.
3. $T_p \to T_0$ **with** $T_p \to T_1$**.** The prefactor $1 - A_0\alpha/D_{0T_p}$ approaches a positive value (since $\alpha < 1/y_1$), so the CC has the sign of ρ (lognormal).

5.4 Sensitivity to ATM slope (skew)

The displaced-diffusion machinery of Lee–Wang (2009) translates the parameters $(a_{\text{swp}}, a_{\text{asw}})$ into the ATM level and slope of the implied volatility smile. Hence (13) effectively prices the ATM skew dependence of the CC. The static-replication trick familiar from CMS does *not* translate easily to ASW-lets because the pricing formula (5) contains *both* $R_{T_0}^{\text{asw}}$ and $R_{T_0}^{\text{swp}}$ in the integrand – a 2-D static replication or an explicit copula is required for full smile fitting.

6 Numerical Sanity Check

Lognormal CC, paper Table 2. With $R_0^{\text{asw}} = 490$bp, $R_0^{\text{swp}} = 4.29\%$, $\sigma_{\text{swp}} = 30\%$, $T_0 \approx 5$Y, schedule 09/02/2016 → 07/02/2021, $T_p = 08/02/2017$, formula (14) gives:

$\sigma_{\text{asw}} \backslash \rho$	−90%	−50%	0%	+50%	+90%
20%	−0.09%	−0.05%	0.00%	+0.06%	+0.12%
30%	−0.13%	−0.08%	0.00%	+0.10%	+0.19%
50%	−0.19%	−0.12%	0.00%	+0.18%	+0.37%

Sensitivity to ASW slope, paper Table 3. Holding $\sigma_{\text{asw}}, \sigma_{\text{swp}}, \rho$ fixed and varying the displacement a_{asw} (i.e. tilting the ATM ASW skew):

$\sigma_{\text{asw}} \backslash \rho$	−90%	−50%	0%	+50%	+90%
20%	0.02%	0.01%	0.00%	0.01%	0.03%
30%	0.05%	0.02%	0.00%	0.02%	0.08%
50%	0.11%	0.04%	0.00%	0.07%	0.28%

The skew impact is small but non-negligible at high vol/correlation levels.

Funding curve (paper Table 5). The benchmark used in the paper is dated 10/02/2011 and ranges out to 14/02/2023; specific discount factors are reproduced in the source PDF. We omit them here.

7 Hedging Considerations (Paper §6)

Hedging a short ASW-let requires trading the underlying *forward asset-swap package* – which embeds a forward position in the bond. Forward repo on government bonds is rarely liquid, but for sizeable sovereign issuers (France, Italy, Belgium, Germany, plus top-tier financial institutions) viable approaches include:

- synthesising forward bond positions via zero-curve estimation / bootstrapping (Bundesbank 1997; BIS 2005);
- Nelson–Siegel (1987) parametric curve construction;
- coupon stripping for liquid sovereign issuers.

The (cross-)gamma cost realised when dynamically buying/selling the forward asset-swap package is precisely what the convexity correction (9) is built to capture.

8 Model Limitations (Paper §8)

8.1 Linear approximation

Assumption 1 can be loose for bonds with long residual life. Alternative functional forms (Hagan 2003, formulas A.3, A.8, A.12a, A.15a) or 2-factor models are available, but they generally require numerical integration and additional parameters that cannot easily be calibrated to liquid market instruments at short notice.

8.2 ASW volatility

σ_{swp} is implied from listed swaptions; σ_{asw} is *not* quoted. Practical sources:

- historical estimation of ASW spread time-series;
- bond-option implied vol when callable bonds with usable surfaces exist;
- credit-spread-vol proxies for issuers with liquid CDS option markets.

8.3 Correlation ρ

Not market quoted – historical estimation is essentially the only practical route. The CC has *linear* sensitivity to ρ in the lognormal case (eq. 14 expanded to first order in $\sigma_{\text{swp}}\sigma_{\text{asw}}\rho T_0$), so ρ uncertainty translates directly into CC uncertainty.

8.4 Skew/smile

Material in stressed regimes: paper Table 3 shows a $\sim 0.3\%$ CC contribution from skew at $\sigma_{\text{asw}} = 50\%$, $|\rho| = 90\%$. The static replication route (Hagan-style) does not trivially extend because of the 2-D nature of the integrand.

9 Validation Notes for the Pricing Library

9.1 Building blocks

1. **Annuity.** Implement (1) with the schedule day-count conventions. Smoke-test: $A_0\, R_0^{\text{swp}} = D_{0T_0} - D_{0T_n}$ (par-swap identity).
2. **Risk-free benchmark bond price.** Construct B_t^{rf} from the bond's promised cashflows on the risk-free curve. Sanity: at issue $B_0^{\text{rf}} \approx 1$ for a par bond.

3. **ASW spread reproduction.** Use (2) on liquid bonds and check the result matches the broker quote (after the usual front-/back-/full-coupon conventions).

9.2 Linear-model calibration

4. Verify $\alpha = 1/\sum y_i$ from Proposition 1.
5. Compute β_U from (7) for each $U \in \{T_p, T_1, \ldots, T_n\}$. Verify the consistency identity $\sum_i y_i\,(\alpha + \beta_{T_i} R_0^{\mathrm{swp}}) = 1$.
6. Bracket-check: at the limit $R_0^{\mathrm{swp}} \to 0$ verify that (8) reduces to $1 = \alpha \sum y_i$.

9.3 Cross-moment

7. Implement (13) and validate against a Monte Carlo of $(R^{\mathrm{swp}}, R^{\mathrm{asw}})$ jointly under (10)–(12) on $[0, T_0]$ with a fine timestep.
8. Limit checks:
   - $a_{\mathrm{swp}} = a_{\mathrm{asw}} = 0$: $\mathbb{E}[R^{\mathrm{swp}} R^{\mathrm{asw}}] = R_0^{\mathrm{swp}} R_0^{\mathrm{asw}} e^{\sigma_{\mathrm{swp}}\sigma_{\mathrm{asw}}\rho T_0}$.
   - $\rho = 0$: $\mathbb{E}[R^{\mathrm{swp}} R^{\mathrm{asw}}] = R_0^{\mathrm{swp}} R_0^{\mathrm{asw}}$ in the lognormal case; $\mathbb{E}[R^{\mathrm{swp}} R^{\mathrm{asw}}] = (R_0^{\mathrm{swp}} + a_{\mathrm{swp}})(R_0^{\mathrm{asw}} + a_{\mathrm{asw}}) - a_{\mathrm{swp}} R_0^{\mathrm{asw}} - a_{\mathrm{asw}} R_0^{\mathrm{swp}} = R_0^{\mathrm{swp}} R_0^{\mathrm{asw}}$ in the displaced case (i.e. no displacement effect when uncorrelated).
   - $\sigma_{\mathrm{swp}} = 0$: $\mathbb{E}[R^{\mathrm{swp}} R^{\mathrm{asw}}] = R_0^{\mathrm{swp}} \mathbb{E}[R_{T_0}^{\mathrm{asw}}] = R_0^{\mathrm{swp}} R_0^{\mathrm{asw}}$.

9.4 Convexity correction

9. **Reduction to no-credit case.** If $B = B^{\mathrm{rf}}$ exactly, $R_t^{\mathrm{asw}} \equiv 0$, $\mathbb{E}[R^{\mathrm{swp}} R^{\mathrm{asw}}] = 0$, $R_0^{\mathrm{asw}} = 0$, and $\mathrm{CC} = 0$ – a trivial regression test.
10. **Lognormal table reproduction.** Reproduce paper Table 2 (Section above) using (14) only – this is the cleanest validation point.
11. **Displaced-vs-lognormal cross-check.** For small $a_{\mathrm{swp}}, a_{\mathrm{asw}}$, the displaced CC should differ from the lognormal CC by $O(a_{\mathrm{swp}}\,\sigma_{\mathrm{asw}}\rho T_0)$ at leading order; verify numerically.

9.5 Full-price validation

12. ASW-let normalised value: $D_{0T_p}\big(R_0^{\mathrm{asw}} + \mathrm{CC}\big)$. Compare against a Monte Carlo computing

$$v_0 \;=\; A_0\,\mathbb{E}^A\Big[R_{T_0}^{\mathrm{asw}}\,G_{T_p}\big(R_{T_0}^{\mathrm{swp}}\big)\Big] \;=\; A_0\,\mathbb{E}^A\Big[R_{T_0}^{\mathrm{asw}}\big(\alpha + \beta_{T_p} R_{T_0}^{\mathrm{swp}}\big)\Big]\,.$$

13. Check the value scales linearly with the coverage factor and the notional.

9.6 Sign and limits

14. For T_p near T_1 (most common case), CC has the sign of ρ in the lognormal regime; verify across the full ρ grid.
15. Check the prefactor $1 - A_0\alpha/D_{0T_p}$ is positive and bounded for all sensible payment-date choices.

9.7 Smile / skew (advanced)

16. For non-flat smile, replicate (13) via

    - displaced-diffusion approximation (1-parameter slope control) – the recommended route per Section 8 of the paper;
    - 2-D static replication using a copula on top of the marginal smiles – heavier but more flexible.

Reference. Pucci, M. (2012a). *Constant Maturity Asset Swap Convexity Correction.* Working paper, Banca IMI, Milan. SSRN no. 1961545. (Subsequently published in summary form in *Risk Magazine*, April 2012.)

Companion papers. M. Pucci (2012b), *Pricing Index-Linked Hybrids under a Local Volatility Market Model*, SSRN 2056277; M. Pucci (2014), *Constant Maturity Treasury Convexity Correction*, IJTAF 17(8), SSRN 3387961; M. Pucci (2019), *Hedging the Treasury Lock*, SSRN 3386521.

Key external references. P.S. Hagan, *Convexity Conundrums: Pricing CMS Swaps, Caps, and Floors*, Wilmott Magazine (2003); A. Pelsser, *Mathematical Foundation of Convexity Correction*, Quantitative Finance 3(1) (2001); P.J. Hunt and J.E. Kennedy, *Financial Derivatives in Theory and Practice*, Wiley (2004), §13.3.3 (linear swap-rate model); R. Lee and D. Wang, *Displaced Lognormal Volatility Skews*, Annals of Finance (2009).